

JEE MAIN REVISION MASTER: MOTION IN A STRAIGHT LINE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 121 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Velocity-Time & Position-Time Graphs	75 PYQs	Graphical Analysis
Very High	Kinematic Equations & Free Fall	46 PYQs	Kinematics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Velocity-Time & Position-Time Graphs	$v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$, $\int v \, dt = \Delta x$
Kinematic Equations & Free Fall	$v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s_n = u + \frac{a}{2}(2n - 1)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Velocity-Time & Position-Time Graphs (Very High Yield — 75 PYQs)

Core Principle: Interpretation of motion curves: slope of x-t is velocity, slope of v-t is acceleration, area under v-t is displacement.

■ **Caution / JEE Pitfall:** Area under speed-time curve gives distance (always positive), while area under velocity-time curve gives displacement.

■ **Speed Trick:** When a is given as a function of x, use $a = v(dv/dx)$ rather than (dv/dt) .

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begi..." [Ans: C]
- [2026 Evening Shift] Q#2 (MCQ): "The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40 s . The total distance travelled b..." [Ans: C]
- [2026 Evening Shift] Q#2 (MCQ): "The velocity (v) - Distance (x) graph is shown in figure. Which graph represents acceleration(a) versus distance (x) variation of th..." [Ans: D]

3.2 Kinematic Equations & Free Fall (Very High Yield — 46 PYQs)

Core Principle: Uniformly accelerated rectilinear motion, vertical motion under gravity, and displacement in the n th second.

■ **Caution / JEE Pitfall:** Assign a consistent sign convention (e.g. upward positive, downward negative) at the beginning of problem solving.

■ **Speed Trick:** Stopping distance $d \propto u^2$. For an object dropped from rest, distance ratio in successive seconds is 1 : 3 : 5 : 7 (Galileo's odd numbers).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A gas balloon is going up with a constant velocity of 10m/s . When this balloon reached a height of 75 m , a stone is dropped from it and ..." **[Ans: D]**
- **[2026 Evening Shift] Q#3 (MCQ):** "A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts deaccelerating with 3m/s^2 . At 10 m heigh..." **[Ans: A]**
- **[2025 Morning Shift] Q#2 (MCQ):** "Which of the following curves possibly represent one-dimensional motion of a particle? Choose the correct answer from the options g..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A gas balloon is going up with a constant velocity of 10m/s . When this balloon reached a height of 75 m , a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is ____ m. (Take $g=10\text{m/s}^2$)

- (A) 85
- (B) 150
- (C) 129
- (D) 125 **[Correct]**

Correct Answer: D

Solution Analysis: When an object is dropped from a moving carrier (like the balloon), it gets the velocity of that carrier at the exact moment of release. Initial velocity of the stone (balloon) is $u(s)=10\text{m/s}$ (upwards). Initial height of the stone is $h=75\text{m}$. The acceleration due to gravity is $g=-10\text{m/s}^2$ (downwards). Using the second equation of motion for the stone's displacement along vertical direction, $y=u(y)t+(1/2)a(y)t^2$ Since the stone eventually hits the ground (at height 0), its displacement from the starting point (75 m) is $y = -75\text{ m}$. $-75=(10)t+(1/2)(-10)t^2$ $\Rightarrow$

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground, at the instant when the first drop strikes the ground is ____ m. ($g=10\text{m/s}^2$)

- (A) 3.8
- (B) 4.0
- (C) 4.2 [Correct]
- (D) 2.5

Correct Answer: C

Solution Analysis: The first drop falls a distance of $H=5\text{m}$ starting from rest ($u=0$). Using the equation of motion: $s=ut+(1/2)gt^2 \Rightarrow 5=0+(1/2) \times 10 \times T^2 \Rightarrow 5=5T^2 \Rightarrow T^2=1 \Rightarrow T=1\text{s}$ When the first drop hits the ground, the sixth drop is just beginning to fall. This means there are 5 equal time intervals between the first and sixth drops. $\Delta t=(T/5)=(1/5)=0.2\text{s}$ At the instant the first drop strikes the ground ($t=1\text{s}$): Drop 1 has fallen for 1.0 s Drop 2 has fallen for 0.8 s Drop 3 has fallen for 0.6 s Drop 4 has fall